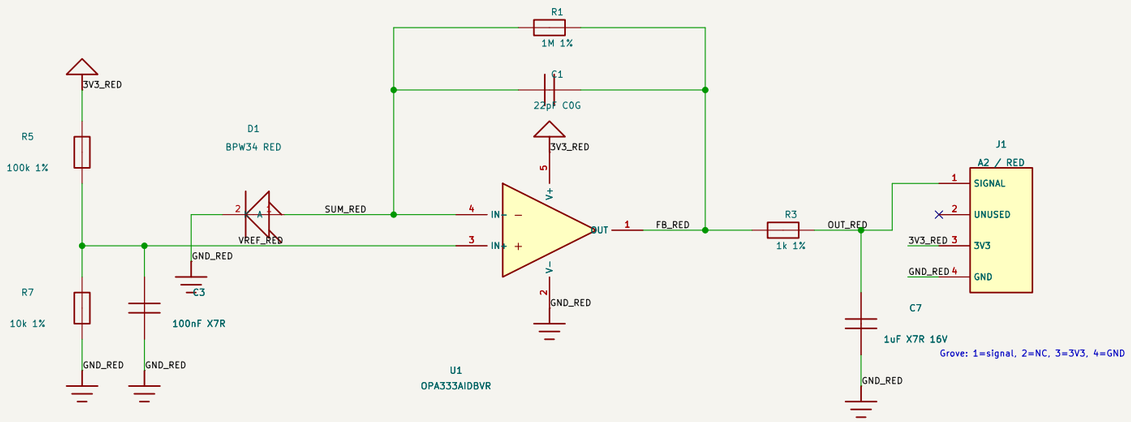
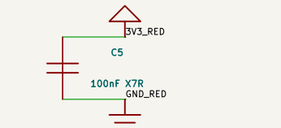
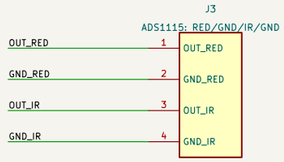
RED / BPW34 -> OPA333 -> A2 / RED



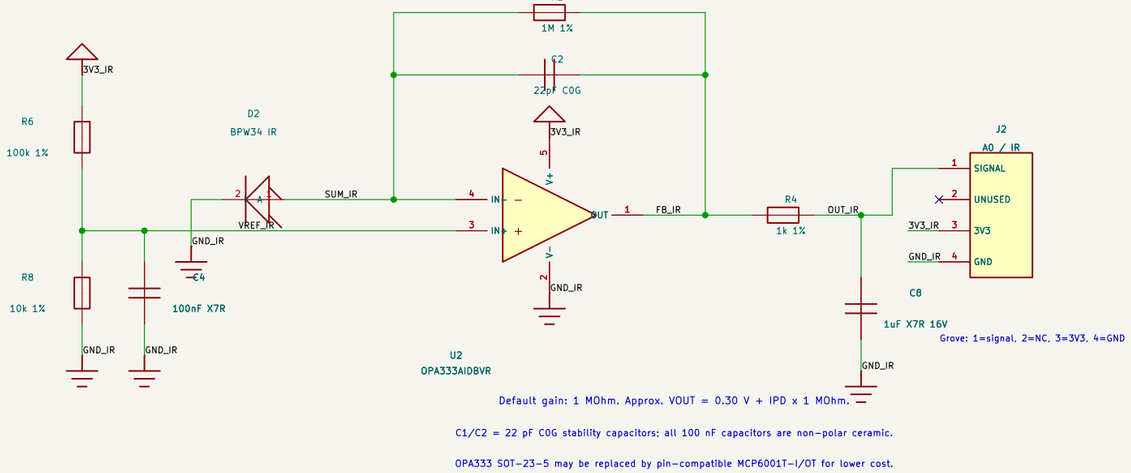


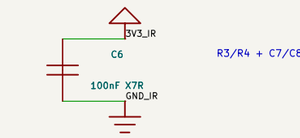


J3 -> ADS1115: each signal travels with its own channel ground

Power the ADS1115 separately. J3 does not provide VCC.

IR / BPW34 -> OPA333 -> A0 / IR





R3/R4 + C7/C8: 1k / 1uF output low-pass, fc = 159 Hz.

Default gain: 1 MOhm. Approx. VOUT = 0.30 V + IPD x 1 MOhm.

C1/C2 = 22 pF C0G stability capacitors; all 100 nF capacitors are non-polar ceramic.

OPA333 SOT-23-5 may be replaced by pin-compatible MCP6001T-I/OT for lower cost.

H1 M3 / 3.2mm NPTH

H2 M3 / 3.2mm NPTH

H3 M3 / 3.2mm NPTH

H4 M3 / 3.2mm NPTH

70 x 30 mm / Grove A2 RED + A0 IR / ADS1115 output

Nguyen Nhat Huy – Pham Thanh Vy

Sheet: /

File: bpw34_receiver.kicad_sch

Title: DATN: PPG-Simulator – Dual BPW34 receiver

Size: A3	Date: 2026-09-29	**Rev: 1.3**
KiCad E.D.A. 10.0.6		Id: 1/1